Miles Choquette

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EDUCATION

Northeastern University Boston, MA
Master of Science, Quantitative Finance | *GPA 3.6* Expected Dec 2026
- Coursework: Fixed Income, Quantitative Portfolio Management, Empirical Methods in Finance, Advanced Topics in Finance

Florida Atlantic University Boca Raton, FL
Bachelor of Science, Computer Science; Certificate in Artificial Intelligence | *GPA 3.45* May 2025

PROJECTS

HVAC Load Calculation Software | *Python* 2026
- Built a tool that reads the vector geometry of CAD-exported architectural PDFs, detects walls, rooms, windows and doors, and passes them to a browser review UI for correction before calculation.
- Implemented ASHRAE Radiant Time Series cooling and steady-state heating loads, sweeping a full design day per room to find its actual peak hour, with ASHRAE 62.1 ventilation and zone-level equipment sizing.
- Added a duct design surface that auto-sizes supply and return runs to required airflow and exports duct schedules, equipment tables and DXF drawings, replacing work previously spread across three programs.

Nelson-Siegel / Diebold-Li Treasury ETF Rotation | *Python, statsmodels* Spring 2026
- Fit Nelson-Siegel to monthly Treasury par curves (2008 to 2024) and forecast level, slope and curvature with an expanding-window VAR(1); combined curve-mispricing and directional signals to allocate across 7 Treasury ETFs.
- Top-3 portfolio earned a Sharpe ratio of 0.39 vs. 0.29 for equal weight, and 1.09 vs. 0.59 over 2018 to 2021; factor attribution showed slope carries the predictive power. Tested significance with Jobson-Korkie.

Polymarket Earnings Signal | *Python* 2026
- Built a no-look-ahead pipeline over 1,231 resolved earnings-beat contracts on ~415 tickers; ran calibration tests, an encompassing regression and an event study against a firm-specific beat-rate baseline.
- Market probabilities beat the baseline (Brier 0.149 vs. 0.185; coefficient 5.78, $p < 10^{-35}$), but permutation, block-bootstrap and Jobson-Korkie tests showed the trading edge was not significant once same-day clustering cut the sample to ~181 independent dates.

Accessible Indoor Navigation (Senior Project) | *Python, AWS Lambda* Spring 2025
- Built the real-time indoor positioning engine for a stadium navigation app, converting Bluetooth signal strength from six beacons to distance and locating the user by minimizing squared range error with L-BFGS-B, smoothed by a Kalman filter.
- Wrote deviation detection that tracks a user against the planned route and triggers recalculation off-track, integrated with AWS Lambda pathfinding for wheelchair, crutches and default mobility profiles.

EXPERIENCE

Adicot Engineering | *Software Engineer & Junior Designer* May 2025 – Present
- Sole developer of the firm's Flask platform for live quote pricing, client intake, job tracking and signed PDF deliverables; designed pricing to fall back to known rates so a bad edit can never produce a wrong quote.
- At a three-person firm, learned new skills on the fly to cover software, HVAC design and client deliverables.

Draper Laboratory | *Student Undergraduate Engineer* May 2024 – May 2025
- Designed an inspection interface and the backend behind it, automating manual data entry and analysis, with potential to cut a 400-hour inspection process to about 40 hours.
- Built a database logging user interactions with the tool to support data-driven process decisions.

360 Huntington Fund | *Analyst, Portfolio & Research* Sep 2025 – 2026
- Pitched Buy recommendations on Waystar (WAY) and JFrog (FROG) to the investment committee; first pitch passed unanimously.
- Rebuilt the portfolio dashboard for the ~$2M, 46-position student-run fund, improving data accuracy and reporting.

SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), SQL, MATLAB, Java, C++, Flask, Git
Quantitative methods: Yield-curve modeling, VAR, Kalman filtering, backtesting, bootstrap and permutation inference, machine learning
Tools: Bloomberg Terminal, Jupyter, Excel